

Exercises for session 2: Discrete time dynamic optimization

1 Optimal fisheries management

We consider an exploited fish stock, the dynamics of which is governed by

$$X_{t+1} = f(X_t) - C_t \quad \text{where} \quad f(x) = 2x - x^2$$

Here, the harvest $C_t \geq 0$ is such that $X_{t+1} \geq 0$ for all t . We aim to choose non-negative $\{C_0, C_1, \dots, C_{N-1}\}$ which maximize

$$J = \sum_{t=0}^{N-1} \sqrt{C_t}$$

subject to the constraint that all X_t are non-negative, and a fixed value for the initial state $X_0 = x_0$.

Question 1: Use your favorite numerical optimizer to solve the problem. Take $X_0 = x_0 = 0.1$. First take $N = 10$; then take $N = 100$. Plot the optimal state sequence $X_0, \dots, X_N$ as well as the harvest $C_0, \dots, C_{N-1}$, both as time series and in the (x, c) -plane.

Hint: The easiest is (probably) to optimize over the states: For a guess on the state sequence X_t , compute the harvest C_t and from that the objective J . Remember that $X_t \geq 0$ and $C_t \geq 0$. You may also use that $X_t \leq 1$ to help the optimizer.

Question 2: We note, when N is large, that the solution is almost constant in the majority of the time. To investigate this, we consider optimal steady-state control:

$$\max\{\sqrt{c} : x, c \geq 0, x = f(x) - c\}.$$

Solve this optimization problem analytically. *Hint:* A sketch in the (x, c) -plane will be helpful. Compare with the numerical solution from the previous.

Question 3: We now add a reward for leaving fish in the ocean for the next generation. Specifically, for given $\psi > 0$, we modify the reward to be maximized to

$$J = \psi X_N + \sum_{t=0}^{N-1} \sqrt{C_t}.$$

Repeat the numerical solution with different values of ψ between 0 and 5. Can you find ψ such that the optimal trajectory ends at the equilibrium point found in question 2? *Note:* Do not spend too much time on this last question; we will consider it in more detail in the following question.

Question 4: Introduce the co-states Λ_t associated with the constraints $X_{t+1} = f(X_t) - C_t$. Write up the co-state equation governing the sequence $\{\Lambda_t\}$, and the optimality condition that couples C_t and Λ_{t+1} . Finally, state the boundary condition on Λ_N . From this, find analytically the value of ψ so that the equilibrium from question 2 also satisfies the dynamic equations of state, co-state, and control.

Question 5: From one of the optimal tracks (X, C) that you computed in the previous, determine the corresponding co-state $\{\Lambda_t\}$ and verify that it satisfies the co-state equation.

Question 6: Experiment with different values of x_0 , and ψ and plot the solutions both as time series and in the (x, c) -plane. Try, for example $N \in \{10, 50, 100\}$, $x_0 \in \{0.01, 0.1, 0.9, 0.99\}$, and $\psi \in \{0, 0.75, 1, 1.5, 2\}$. Does the plot in the (x, c) -plane remind you of a particular class of dynamic systems?

Question 7: (Optional) We now include *discounting* in the reward, i.e., the objective function is modified to

$$J = \sum_{t=0}^{T-1} r^t \sqrt{C_t}$$

where $r = 0.95$. Solve the optimization problem as in question 1 numerically and compare the solution to the one without discounting.

Question 8: (Optional) Compute analytically the equilibrium of the optimal system with discounting. How does the steady-state value of the biomass x and the catch c depend on the discounting rate r ?